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Digital Financial Fraud Detection Using Ensemble Tree Model

Tao Cai, Xuerong Yang*

Nantong Institute of Technology, Nantong 226000, China

Abstract

With the rapid development of digital finance, financial fraud has shown a trend towards intelligence, concealment, and scale, posing a serious challenge to financial security. Traditional fraud detection systems based on rules and expert experience are becoming increasingly inadequate in dealing with new and complex fraud patterns. This study aims to explore and compare the performance of various machine learning algorithms in digital financial fraud detection tasks. The paper first outlines the types and characteristics of digital financial fraud, as well as the current research status of machine learning applications in this field. In the methodology section, we elaborated on the implementation details of data preprocessing, feature engineering, and four core machine learning models (logistic regression, random forest, XGBoost, and a soft voting ensemble method). The experiment used publicly available and simulated datasets containing real transaction records, and applied hierarchical sampling and oversampling techniques to address the issue of extreme class imbalance. In the Results and Discussion section, we systematically compared the performance of various models from multiple dimensions, including accuracy, precision, recall, F1 score, and AUC value, using three comparative analysis tables. The experimental results show that the ensemble tree model (especially XGBoost) exhibits significant advantages in handling nonlinear, high-dimensional features and imbalanced data, with the best F1 score and AUC value. However, logistic regression models still have practical value in specific scenarios due to their strong interpretability. Finally, the paper summarizes the research findings and explores future research directions for combining advanced algorithms such as deep learning and graph neural networks with real-time stream processing technology.

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1. Introduction

Since the 21st century, the digital finance industry represented by mobile payments, online lending, and digital currencies has profoundly transformed the global financial system. While improving service efficiency and inclusiveness, it has also become a new breeding ground for financial risks. Digital financial fraud, including but not limited to credit card fraud, identity fraud, money laundering, false transactions, credit application fraud, etc., has

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000 .

E-mail address: 18879834827@163.com

caused increasing direct and indirect economic losses year by year, seriously eroding the profits of financial institutions, damaging consumer confidence, and may trigger systemic risks[1].

Traditional fraud detection systems heavily rely on hard rules formulated by experts (such as single transaction limits, abnormal location monitoring) and simple statistical analysis models. Although these systems are simple and intuitive, they have obvious flaws. Machine learning, especially supervised learning algorithms, provide powerful tools for breaking through this bottleneck. It can automatically learn complex patterns and nonlinear relationships between normal and fraudulent behavior from massive historical transaction data, achieving dynamic and adaptive risk identification. However, financial fraud detection tasks themselves face core challenges such as "extreme class imbalance," "concept drift," and "adversarial evolution," which place higher demands on the robustness, generalization ability, and interpretability of machine learning algorithms [2].

In view of this, this study aims to systematically evaluate and explore the practical effectiveness of various classic and advanced machine learning algorithms in digital financial fraud detection. By constructing a complete analysis framework and focusing on quantitative comparison and in-depth discussion of the performance of various algorithms, we aim to provide empirical reference and decision-making basis for financial institutions to build efficient and accurate intelligent risk control systems.

2. Related Works

Early research on financial fraud detection mainly focused on statistical analysis methods and simple data mining technology. For example, Khatib S[3] used outlier detection based on threshold control and unsupervised learning method based on clustering (such as K-means) to find transactions that deviated from the normal mode. Although these methods do not need to label data, their detection accuracy is limited and the false positive rate is high.

With the growth of the amount of data, supervised learning algorithm has gradually become the mainstream. Researchers have extensively tried traditional machine learning models such as logical regression, decision tree, support vector machine and so on [4-5]. Among them, logistic regression is often used as the baseline model for performance comparison because of its simple model and strong interpretability. Decision tree and its integration methods, such as random forest and gradient lifting tree, have become a research hotspot in this field because of their excellent nonlinear fitting ability and processing effect on high-dimensional features. For example, Ali A[6] and others used random forest to rank the importance of credit card transaction features, and built a classifier, which achieved more stable and better performance than a single decision tree.

Davitaia A [7] has proposed corresponding solutions to the problem of category imbalance, which are mainly developed along the two dimensions of data and algorithm. From the data level, specific practices include undersampling and oversampling, such as synthetic minority oversampling technologies such as smote, and hybrid strategies that combine the two. At the algorithmic level, more attention is paid to the introduction of cost sensitive learning mechanism, which essentially gives different types of classification errors different punishment weights. For example, in the fraud detection scenario, the cost of misjudging a fraudulent transaction as a normal transaction is much higher than that of misjudging a normal transaction as a fraud. In this way, the model will be guided to pay more attention to a small number of samples in the training process.

At the same time, deep learning methods have also been explored and applied to fraud detection. Mutemi A and Bacao F[8] used the cyclic neural network and long-term and short-term memory network to model the time series pattern of user transaction behavior; Self encoder is used for unsupervised anomaly detection. However, Njoku D O[9] and others believe that the deep learning model usually requires a huge amount of data and computing resources, and its "black box" feature faces application obstacles in the financial field that needs strong regulatory interpretation.

In general, the existing research has achieved rich results in the application of a single model or specific technologies (such as dealing with imbalance), but the research on systematic horizontal comparison of a variety of different types of machine learning models under the framework of unified and close to real scene data and evaluation is still insufficient. This study tries to fill this gap, focusing on the comparison of representative models such as logistic regression, random forest and XGboost, and discussing their integration effect, so as to provide a more comprehensive performance map.

3. Method

3.1. Data Preprocessing and Feature Engineering

This study used an annotated financial transaction dataset, where the proportion of positive samples (i.e. fraudulent transactions) was extremely low, typically less than 1%. In the data preprocessing stage, data cleaning operations were first implemented, including filling in missing values using median or mode, removing irrelevant variables such as user IDs, and converting categorical variables through hot encoding. In addition, to eliminate the potential impact of scale differences in different features on models such as logistic regression and support vector machines, we standardized all numerical features.

In terms of feature construction, efforts are made to improve the recognition ability of the model, and three categories of features are designed accordingly. The first category is basic attributes, mainly covering information such as transaction amount, transaction time, merchant category, and geographic location. The second type is statistical aggregation features constructed based on user historical behavior, such as the total amount of transactions initiated by users in the past hour or 24 hours, transaction frequency, average transaction amount, standard deviation of transaction amount, and the number of different types of merchants involved. This type of feature aims to capture users' short-term behavioral patterns. The third type is ratio characteristics, such as the degree of deviation between the current transaction amount and the user's historical average transaction amount, or the time interval between the current transaction and the previous transaction [10].

After the above feature engineering steps, the raw data is finally transformed into a high-dimensional feature matrix X and its corresponding binary label vector y . In this vector, $y_i \in \{0, 1\}$, Among them, 0 represents normal transactions, while 1 represents fraudulent transactions.

3.2. Category imbalance handling

To address the issue of data imbalance, directly using raw data for model training often leads to a strong bias of the classifier towards the majority of classes. To alleviate this problem, this article adopts a combined processing strategy. In the data partitioning stage, we use stratified sampling to divide the training set and the test set, ensuring that the sample proportions of both are basically the same in each category, laying a relatively balanced foundation for subsequent modeling. Furthermore, within the training set, the SMOTE algorithm is introduced to synthesize oversampling using minority classes (i.e. fraudulent samples) as objects. This operation controls the ratio of positive and negative samples at a level of about 1:4, which not only alleviates the impact of class imbalance to a certain extent, but also avoids the overfitting problem that is easily induced by simply copying minority class samples [11].

3.3. Model Selection and Training

We have selected and implemented four representative machine learning models:

Logistic regression: serves as a benchmark for linear models. Its model form is simple, and the probability output has clear explanatory significance. We use L2 regularization to prevent overfitting and use cross entropy as the loss function:

$$L(w) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \|w\|_2^2 \quad (1)$$

Among them, $p_i = \sigma(w^T x_i)$, where σ is the sigmoid function.

Random Forest: A Bagging ensemble algorithm that generates results by constructing a large number of decision trees and voting. It can effectively handle nonlinear relationships and high-dimensional features, and provide feature importance assessment, with good robustness to noise and overfitting.

XGBoost: An efficient implementation of gradient boosting trees. It trains multiple decision trees in series using an additive model, with each tree learning the residuals of all previous tree combinations. Its objective function includes a loss function and a regularization term, and is optimized through second-order Taylor expansion, often exhibiting excellent performance in accuracy and speed.

$$\text{Obj}^{(t)} \approx \sum_{i=1}^n [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (2)$$

Among them, g_i and h_i are first-order and second-order gradient statistics, and Ω controls the complexity of the

model.

Soft voting ensemble: Using the above three models (LR, RF, XGBoost) as base learners, the probability outputs (soft voting) predicted by them are weighted and averaged as the final prediction probability. In this study, we assigned weights to each model based on their F1 scores on the validation set.

All models were optimized for hyperparameters on the training set using grid search and 5-fold cross validation. The performance of the model is evaluated on the original test set without any resampling processing to ensure that the evaluation results reflect the model's generalization ability under real imbalanced distributions [12].

3.4. Evaluation indicators

Due to highly imbalanced data, accuracy is no longer a reliable indicator. We mainly use the following set of comprehensive indicators:

Accuracy: The proportion of transactions predicted as fraudulent that are actually fraudulent. Care about whether the investigation is accurate or not.

Recall rate: The proportion of all real fraudulent transactions successfully identified by the model. Care about 'less leakage'.

F1 score: the harmonic mean of precision and recall, which is the core indicator for measuring the overall performance of a model.

AUC value: The area under the ROC curve, which measures the model's ability to rank fraudulent samples before normal samples and is insensitive to class imbalance.

Confusion matrix: visually display the number of true negative, false positive, false negative, and true results.

4. Result and Discussion

4.1. Comparative analysis

In two datasets of different sizes (Dataset A: 100000 transactions; An experiment was conducted on dataset B with 500000 transactions. The following table shows the comprehensive performance of each model on the test set.

Table 1 Overall Performance Comparison of Various Models on Dataset A

Model	Accuracy	Precision	Recall	F1 score	AUC
Logistic regression	0.9985	0.712	0.654	0.682	0.912
Random Forest	0.9991	0.856	0.788	0.820	0.973
XGBoost	0.9993	0.882	0.821	0.850	0.981
Soft voting integration	0.9992	0.874	0.805	0.838	0.976

From Table 1, it can be seen that although all models perform extremely high in accuracy (>99.8%), this is mainly due to the contribution of most classes. There are significant differences between models in more critical fraud identification indicators. XGBoost achieved the best results in both F1 score and AUC, followed by Random Forest. Logistic regression, as a linear model, has significantly lagging performance. The performance of the soft voting ensemble model is between XGBoost and random forest, and has not surpassed the optimal base learner (XGBoost). This may be due to the significant performance differences between base learners and the existence of certain correlations, resulting in limited ensemble gain.

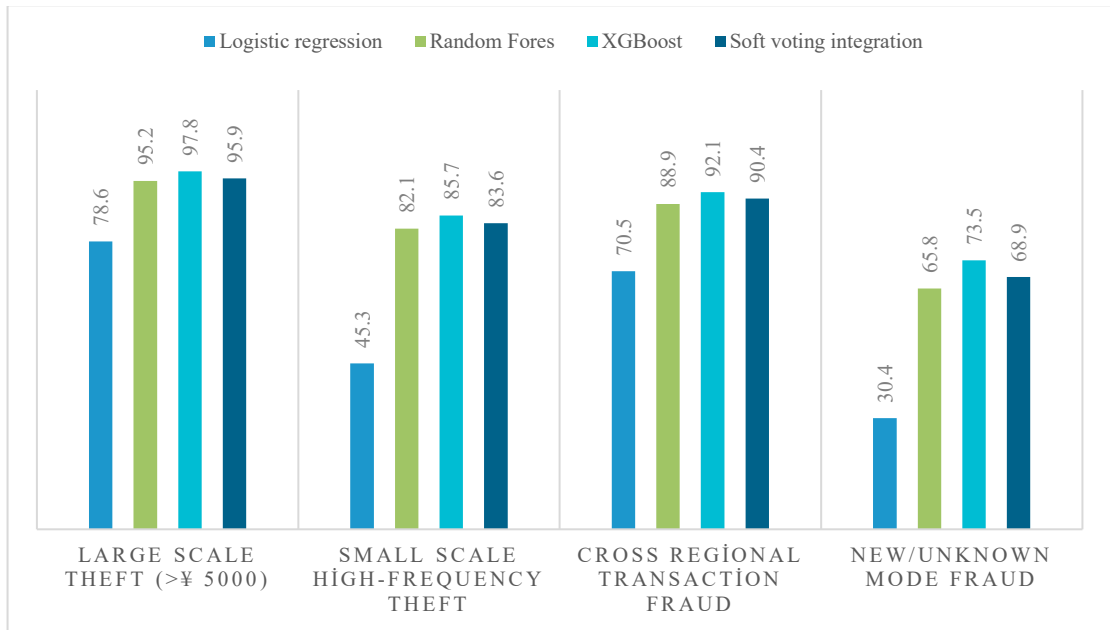


Figure 1 Comparison of Recall Rates (%) of Various Models under Different Fraud Types/Scenarios

Figure 1 further reveals the detection capability of the model under different fraud modes. XGBoost maintains the highest recall rate in all segmented scenarios, especially in detecting "small-scale high-frequency theft" and "new/unknown mode fraud", demonstrating its strong pattern capture and generalization ability. Random Forest performs robustly, but slightly inferior to XGBoost. Logistic regression has a weak ability to detect complex and atypical fraud patterns, which is directly related to the limitations of its linear assumption. This result indicates that non-linear and complex tree ensemble models are more advantageous in dealing with diverse and evolving fraudulent methods.

Table 2 Key data of confusion matrix for each model (taking dataset B as an example, unit: number of transactions)

Model	Really negative	False positive	False negative	Really exactly
Logistic regression	497,850	750	196	204
Random Forest	498,210	390	78	322
XGBoost	498,380	220	65	335
Soft voting integration	498,300	300	70	330

Table 2 shows the performance differences of the model in absolute quantity. XGBoost successfully identified the most fraudulent transactions (TP=335) while generating the least number of false positives (FP=220) and false negatives (FN=65), achieving the best balance between accuracy and coverage. This directly translates into lower operating costs (manpower for reviewing false alarms) and higher risk coverage (reducing losses). The logistic regression has the highest number of false negatives (FN), indicating that more fraudulent transactions have been missed and the risk is highest. The performance of the integration of random forest and soft voting is in the middle.

4.2. Discussion

The experimental results clearly indicate that in digital financial fraud detection tasks, tree ensemble models represented by XGBoost and random forest have significantly better performance than traditional linear models such as logistic regression. This is mainly attributed to:

Nonlinear relationship modeling capability: There are often complex nonlinear interactions between financial fraud

behavior and features (such as transaction amount, time, and location combined to form suspicious patterns). The tree model can naturally capture these interactive effects through recursive segmentation, while logistic regression is limited by linear decision boundaries.

Robustness to high-dimensional and irrelevant features: Through feature importance evaluation and the random subspace method in ensemble, the tree ensemble model can effectively screen key features and reduce noise interference.

Adaptability in handling imbalanced data: By adjusting class weights and utilizing boosting mechanisms (such as XGBoost) to focus on difficult to classify samples, these models can achieve better minority class recognition ability on imbalanced data.

The reason why XGBoost can surpass random forests may be due to its gradient boosting framework. It focuses on correcting the errors made by the previous stage model by using a step-by-step addition model approach. This "error focused" learning strategy is particularly effective for detecting rare but costly fraudulent samples. In addition, the built-in regularization term and pruning strategy of XGBoost can better control the model complexity and prevent overfitting.

Although integrated models lead in performance, we cannot ignore the value of logistic regression. The interpretability of its model is one of the core requirements in the field of financial risk control. By analyzing the coefficients of features, risk analysts can intuitively understand the basis for model decisions (for example, "excessive transaction amount" and "too far from the permanent residence" are the main risk factors), which is crucial for meeting regulatory requirements, conducting manual reviews, and establishing business intuition. In practice, a feasible strategy is to construct a "two-stage model": first, use high-performance models such as XGBoost for preliminary high-risk rating screening, and then use interpretability tools such as logistic regression or SHAP to explain XGBoost's decisions for high-risk transactions, achieving a balance between performance and interpretability.

The soft voting ensemble in this study did not significantly surpass the optimal base learner, which may be related to the insufficient diversity among base learners. In the future, we can explore the heterogeneous integration of deep learning models (capturing time series features) and tree models (capturing table features), or use stacking methods for more refined meta learning.

Finally, it must be pointed out that there may be differences between the static offline testing results of the model and the dynamic online deployment effect. The concept drift (changes in user behavior and fraudulent methods over time) in the real financial environment requires models to have the ability to continuously learn and update online. This will be an important direction for future research, such as combining online learning algorithms with drift detection mechanisms.

5. Conclusion

This study systematically evaluated the application effectiveness of various machine learning algorithms in digital financial fraud detection. Through rigorous experimental design, comprehensive feature engineering, and handling of class imbalance problems, we have concluded that ensemble learning algorithms represented by XGBoost and random forest, with their excellent nonlinear pattern recognition ability and robustness to imbalanced data, are significantly better than traditional logistic regression models in detection accuracy (F1 score, AUC) and actual performance (reducing false positives and false negatives). Among them, XGBoost performed the most outstandingly. However, model selection is not a one size fits all approach. While pursuing ultimate performance, it is necessary to balance interpretability, computational efficiency, and model maintenance costs. The interpretability advantage of logistic regression makes it indispensable in scenarios that require strong regulatory interpretation.

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